

Alec Pagliarussi

Staff Software Engineer · Distributed Systems, Platform Architecture, Technical Leadership
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SUMMARY

Staff Software Engineer with 5+ years building and operating distributed backend systems across Schedule Engine, Scheduling Pro, and Contact Center Pro (HomeX, then ServiceTitan). Technical lead of a product squad and architect of the shared SDKs, caching layers, and billing integrations that other teams build on across a multi-squad microservice platform. Focus areas: resilience under dependency failure, turning one-off fixes into reusable foundations, and growing engineers into technical owners.

CORE SKILLS

Languages	TypeScript / JavaScript (Node.js), Java, Python, C#, SQL
Architecture	Microservices, REST and GraphQL API design, shared SDK and package design, event-driven systems, caching and graceful degradation, idempotent integrations, authentication and authorization (SSO, token services, Keycloak)
Data & messaging	MongoDB, PostgreSQL, Redis, Snowflake, Apache Kafka, Azure Service Bus
Cloud & operations	Kubernetes, Docker, Azure, Oracle Cloud, AWS, Grafana, Prometheus, Kibana, release management, on-call and incident response, end-to-end test automation
Frontend	React, backend-for-frontend (BFF) patterns
AI-assisted engineering	Agentic coding workflows, spec-driven development, reusable agent skills for design docs and diagrams

EXPERIENCE

Staff Software Engineer

ServiceTitan — Contact Center Pro (CCP) · Remote, Ontario, Canada

May 2025 – Present

- Technical lead for a CCP product squad** (cloud contact-center platform): set technical direction, run planning and design reviews, and develop engineers into initiative owners who write the design doc and lead delivery. Squad spans Voice Intelligence, Virtual Agent, billing, and reporting, with two years of on-time quarterly roadmap delivery.
- Architected shared foundations adopted by every CCP microservice:** a shared authorization and policy package, a telephony client SDK with built-in caching and real-time multi-cloud routing, and the platform's first shared service-client SDK, which became the standard client pattern across the platform.
- Led the caching and graceful-degradation strategy** for CCP's critical upstream dependencies: 95–98% cache hit rate, 6x fewer outbound calls overall and 63x fewer to the hottest endpoints, with services continuing to serve from stale data through upstream outages. Covered Redis sizing, staged rollout, Grafana observability, and measured validation; owned services ran a full year without a major incident.
- Designed and delivered usage-based billing for Virtual Agent** against the company's internal licensing and metering APIs: idempotent, fault-tolerant, and fully instrumented, with two pre-release defects caught during rollout. Authored the billing architecture roadmap for the product.
- Shipped event-driven integrations on Kafka and Azure Service Bus** (call-agent status reporting, automatic phone-number provisioning on tenant go-live) that became the team's standard pattern for event consumers.
- Re-architected user synchronization** from full delete-and-reinsert to incremental patch updates with a lightweight sync path, cutting downstream load and unnecessary writes and adding step-level observability.
- Built the tenant health-check framework** (author, evaluate, schedule, persist, report) that let several engineers ship checks in parallel instead of serializing behind one owner.
- Led AI-assisted engineering adoption** across CCP and other ServiceTitan teams: spec-driven workflows, reusable agent skills for design docs and diagrams, and a shared knowledge base; presented at the company-wide AI tools demo and delivered an org-wide tech talk on microservice authentication patterns.
- Standardized releases across squads: shared end-to-end test helpers, deployment and rollback runbooks, caching-health dashboards, and analytics pipelines to Snowflake; coordinated the team's remediation of a public npm supply-chain advisory across multiple services.

Senior Software Engineer

Aug 2022 – Apr 2025

ServiceTitan — Schedule Engine, Scheduling Pro, Contact Center Pro · Remote, Ontario, Canada

- **Became technical lead of the CCP product squad in late 2024**, leading design, delivery, and test-automation standards for the team's features while continuing hands-on development.
- **Co-led the migration of 30+ production services from AWS ECS to Kubernetes** with minimal downtime; owned MongoDB cluster partitioning and delivered the MongoDB 7 upgrade across all instances.
- **Designed and delivered Kafka topic partitioning** to scale messaging throughput within the platform's constraints, from proof-of-concept demo through production rollout.
- **Built the scheduler and availability services and the contractor and scheduler APIs and frontends for Scheduling Pro**, the next-generation scheduling product, including a Reserve with Google refactor for backward compatibility.
- **Led the design of the public WorkConnect API** for partner integrations, built the auto-provisioning service and backfill scripts for a major platform launch, and led the WorkConnect authentication refactor to SSO via a token server, sharply reducing access-related support tickets.
- **Led CCP Workflow Management and the Holiday Hours feature**: designed a facade API over the underlying telephony platform, planned the data migration with backward compatibility, and shipped on time for year-end with zero post-launch issues.
- Led Virtual Agent configuration UI and workflow module, personal and group voicemail, and an availability-request logging solution that lets support diagnose scheduling issues without engineering involvement.
- Owned production support for Schedule Engine for a year: resolved the escalation backlog, drove code coverage to 80% across core repositories, and instrumented services with Grafana and Prometheus.
- Ran technical interviews across levels and coached senior engineers to run their own; made test automation part of the squad's definition of done.

Software Engineer I / II

Apr 2021 – Aug 2022

HomeX — Schedule Engine (acquired by ServiceTitan, Aug 2022) · Toronto, Canada

- Full-stack engineer on Schedule Engine, an online scheduling platform for home-services contractors: TypeScript/Node.js services and APIs, GraphQL, React, PostgreSQL and MongoDB.
- Resolved dozens of critical customer-facing production issues in the first three months on the support rotation and added metrics and Grafana dashboards to critical services.

EARLIER EXPERIENCE

- **Software Engineer (contract), George Brown College Research & Innovation, 2020 – 2021**: led a small team building a web application from zero to production with real-time WebSocket communication and role-based authorization (React, Node.js, GraphQL, MongoDB, Redis, Twilio).
- **Volunteer Software Engineer, Working Women Community Center, Toronto, 2020 – 2022**: built a tutoring-program management application (scheduling, group creation, data collection, real-time reporting; TypeScript, React, Node.js, GraphQL, PostgreSQL).
- **Tutor and Teaching Assistant, George Brown College, 2019 – 2021; Robotics Teacher, Jean Piaget College (Brazil), 2015 – 2018; founder of a 3D-printing shop, 2016 – 2018.**

EDUCATION & AWARDS

Advanced Diploma, Computer Programmer Analyst — George Brown College, Toronto

2018 – 2021

- Computer Technology Departmental Award for academic performance and engagement in the college community.
- IEEEExtreme competitive programming: 11th in Canada, top 12% worldwide (2020); 20th in Canada (2019).